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Studierichting: Informatica
Oefeningenreeks 3D nr. 5

$$f(x) = \frac{3x^2 + ax + 6}{3x^2 + bx + 9}$$

$$f'(x) = \frac{(3x^2 + ax + 6)' \cdot (3x^2 + bx + 9) - (3x^2 + ax + 6) \cdot (3x^2 + bx + 9)'}{(3x^2 + bx + 9)^2} =$$

$$= \frac{(6x + a) \cdot (3x^2 + bx + 9) - (3x^2 + ax + 6) \cdot (6x + b)}{(3x^2 + bx + 9)^2}$$

$f'(x) = 0$, dan:

$$\frac{(6x + a) \cdot (3x^2 + bx + 9) - (3x^2 + ax + 6) \cdot (6x + b)}{(3x^2 + bx + 9)^2} = 0$$

$$(6x + a) \cdot (3x^2 + bx + 9) - (3x^2 + ax + 6) \cdot (6x + b) = 0$$

Als $x + 1$, dan:

$$(6 \cdot 1 + a) \cdot (3 \cdot 1^2 + b \cdot 1 + 9) - (3 \cdot 1^2 + a \cdot 1 + 6) \cdot (6 \cdot 1 + b) = 0$$

$$\Leftrightarrow (6 + a)(3 + b + 9) - (3 + a + 6)(6 + b) = 0$$

$$\Leftrightarrow (6 + a)(12 + b) - (9 + a)(6 + b) = 0$$

$$\Leftrightarrow (72 + 6b + 12a + ab) - (54 + 9b + 6a + ab) = 0$$

$$\Leftrightarrow 72 + 6b + 12a + ab - 54 - 9b - 6a - ab = 0$$

$$\Leftrightarrow 18 + 6a - 3b = 0$$

$$\Leftrightarrow 6 + 2a - b = 0$$

$$\Leftrightarrow 2a - b = -6$$

Als $x + 2$, dan:

$$(6 \cdot 2 + a) \cdot (3 \cdot 2^2 + b \cdot 2 + 9) - (3 \cdot 2^2 + a \cdot 2 + 6) \cdot (6 \cdot 2 + b) = 0$$

$$\Leftrightarrow (12 + a)(12 + 2b + 9) - (12 + 2a + 6)(12 + b) = 0$$

$$\Leftrightarrow (12 + a)(21 + 2b) - (18 + 2a)(12 + b) = 0$$

$$\Leftrightarrow (252 + 24b + 21a + 2ab) - (216 + 18b + 24a + 2ab) = 0$$

$$\Leftrightarrow 252 + 24b + 21a + 2ab - 216 - 18b - 24a - 2ab) = 0$$

$\Leftrightarrow 36 - 3a + 6b = 0$

$\Leftrightarrow 3a - 6b = 36$

**Stap 3: Stelsel:**

$$\begin{cases} 2a - b = -6 \\ 3a - 6b = 36 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ 3a - (6 + 2a) \cdot 6 = 36 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ 3a - 36 - 12a = 36 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ -9a = 72 \end{cases}$$

$$\begin{cases} b = 6 + 2a \\ a = -8 \end{cases}$$

$$\begin{cases} b = 6 + 2 \cdot (-8) \\ a = -8 \end{cases}$$

$$\begin{cases} b = -10 \\ a = -8 \end{cases}$$

**Antwoord:** $a = -8$; $b = -10$.

# References